

# Elementary Operations on Quasi-Toeplitz Matrices

## PCCA6

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- Operations on large structured matrices are often limited by memory bandwidth.
- Standard operations on  $n^2$  elements lead to unnecessary computations.

## Goal

Optimize memory footprint and execution time on structured matrices by implementing displacement operators and generator-based representations.

## Where do we see these structured matrices?

- Sylvester matrix
- Diagonal gradients
- Quantum channel calculations in classical quantum simulations
- ...

## Definition

A matrix  $A \in \mathbb{K}^{n \times m}$  is **Toeplitz** if  $A_{i,j} = A_{i+1,j+1}$  for all valid  $i, j$ .

$$A = \begin{pmatrix} a_0 & a_{-1} & \cdots & a_{-(m-1)} \\ a_1 & a_0 & \cdots & a_{-(m-2)} \\ \vdots & \ddots & \ddots & \vdots \\ a_{n-1} & \cdots & a_1 & a_{n-m} \end{pmatrix}$$

# Displacement & Generators

## Displacement Operator

$$\nabla A = A - ZAZ^T = GH^T$$

For Toeplitz:  $\text{rank}(\nabla A) \leq 2$

### Notice

Storing  $(G, H)$  instead of  $A$  reduces memory from  $O(n^2)$  to  $O(\alpha n)$ .

## Reconstruction

$$A = \sum_{k=1}^{\alpha} L(g_k) U(h_k)$$

## Example - Displacement &amp; Generators

Toeplitz Matrix  $A$ 

$$A = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 5 & 4 & 3 & 2 \\ 6 & 5 & 4 & 3 \\ 7 & 6 & 5 & 4 \end{pmatrix}$$

Displacement  $\nabla A = A - ZAZ^T$ 

$$\nabla A = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 5 & 0 & 0 & 0 \\ 6 & 0 & 0 & 0 \\ 7 & 0 & 0 & 0 \end{pmatrix}$$

Only first row and column stays.  $\text{rank} = 2$ .

Thank you